

Depreciation Coverage Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Depreciation Coverage Moderation (DCM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCM achieves an annualized gross (net) Sharpe ratio of 0.42 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (12) bps/month with a t-statistic of 2.44 (1.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets) is 14 bps/month with a t-statistic of 2.01.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding return differences, empirical identification of firm characteristics that capture consumption risk exposure remains challenging. The disconnect between consumption-based theory and observable firm characteristics represents a fundamental puzzle in asset pricing, as most documented return predictors lack clear connections to households' intertemporal marginal rate of substitution or state-dependent consumption dynamics.

We propose that Depreciation Coverage Moderation (DCM) captures firms' differential exposure to consumption risk through its reflection of capital durability and replacement flexibility during economic downturns. Firms with moderate depreciation coverage maintain optimal capital vintage distributions that provide consumption-hedging properties during adverse macroeconomic states (Campbell and Cochrane, 1999). This moderation principle aligns with long-run risk models where agents price assets based on their covariance with persistent consumption growth shocks (Bansal and Yaron, 2004). The depreciation-consumption nexus operates through the production channel: firms with extreme depreciation coverage patterns exhibit heightened sensitivity to aggregate productivity shocks that directly impact households' consumption opportunities.

The state-dependent nature of depreciation coverage effects manifests most strongly during economic contractions when the marginal utility of consumption peaks. Following the logic of (Cochrane, 1991), production-based characteristics like depreciation patterns should price assets to the extent they correlate with the stochastic discount factor. Moderate depreciation coverage signals firms' ability to smooth capital services across states, reducing their covariance with consumption disasters

and lowering required returns (Barro, 2006). This mechanism generates predictable return variation as investors demand compensation for holding firms whose depreciation patterns amplify consumption risk exposure.

The intertemporal dimension of depreciation coverage connects to habit formation models where risk premia vary with the consumption surplus ratio (Campbell and Cochrane, 1999). Firms maintaining moderate depreciation coverage effectively manage the timing option embedded in capital replacement decisions, allowing them to defer major capital expenditures during periods of high marginal utility. This operational flexibility translates into lower systematic risk as these firms' cash flows exhibit reduced correlation with the pricing kernel during consumption disasters (Gabaix, 2012). The resulting cross-sectional return predictability emerges from rational pricing of heterogeneous consumption risk exposures embedded in firms' depreciation policies.

The DCM strategy generated a value-weighted long/short return of 21 basis points per month with a t-statistic of 3.04, yielding an annualized gross Sharpe ratio of 0.42 over the 1973 to 2024 sample period. The strategy's alpha relative to the Fama-French six-factor model equaled 17 basis points monthly with a t-statistic of 2.44, demonstrating robust performance after controlling for established risk factors. Economic significance persisted across size quintiles, with the strategy earning 32 basis points monthly among the largest stocks (t-statistic of 3.51), confirming that DCM's predictive power extends beyond small-cap segments where limits to arbitrage might generate spurious patterns.

Net of transaction costs, the DCM strategy maintained economic viability with a net Sharpe ratio of 0.31 and a generalized alpha of 12 basis points relative to the six-factor model (t-statistic of 1.67). The strategy loaded significantly on the CMA factor with a coefficient of 0.25 (t-statistic of 5.24), consistent with DCM capturing variation in firms' investment patterns that correlate with consumption

risk. Alternative portfolio construction methods confirmed robustness, with nineteen of twenty-five specifications generating t-statistics exceeding three for gross alphas, and eighteen specifications significantly expanding the mean-variance frontier after accounting for implementation costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, DCM retained an alpha of 14 basis points monthly with a t-statistic of 2.01. The strategy’s gross Sharpe ratio of 0.42 exceeded 83% of anomalies in the factor zoo, while its net Sharpe ratio of 0.31 surpassed 91% of documented strategies. Among combination strategies incorporating 155 anomalies from the factor zoo, adding DCM increased terminal wealth from \$993.77 to \$1194.95 using average rank methodology, demonstrating meaningful incremental contribution to the cross-section of expected returns.

This paper extends the consumption-based asset pricing literature by identifying depreciation coverage moderation as a characteristic that captures cross-sectional variation in consumption risk exposure. While (Fama and French, 2015) document that investment and profitability factors help explain average returns, and (Hou et al., 2015) develop a q-factor model based on investment theory, our work provides the first direct link between depreciation patterns and consumption-based pricing kernels. Unlike purely empirical exercises that mine firm characteristics for return predictability, we ground DCM in the theoretical framework of state-dependent marginal utility and demonstrate its connection to systematic consumption risk through production channels.

Methodologically, we advance the anomaly evaluation literature by implementing the comprehensive testing protocol of (Novy-Marx and Velikov, 2023), providing transparent assessment of DCM’s incremental contribution to the factor zoo. Our spanning tests against 209 existing anomalies and robustness to alternative portfolio constructions exceed the rigor of typical factor discoveries, addressing concerns raised

by (Harvey et al., 2016) about multiple testing and data mining in cross-sectional return predictability research. The positive net generalized alphas across multiple factor models confirm that DCM expands the achievable investment opportunity set even after accounting for implementation costs, distinguishing our contribution from paper profits that cannot be realized in practice.

The broader implications extend to understanding how firm-level operational decisions embed consumption risk that rational investors price in equilibrium. By connecting depreciation policies to state-dependent marginal utility through production-based channels, we provide empirical support for consumption-based models while offering practitioners a implementable strategy that captures this risk-return trade-off. Future research can build on our framework to explore how other accounting measures of capital management relate to households’ intertemporal consumption decisions, potentially unifying micro-level firm characteristics with macro-level consumption dynamics in a coherent asset pricing framework.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and depreciation expenses across firms. construction of our Depreciation Coverage Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the total proceeds from the issuance of long-term debt instruments during the fiscal year, capturing firms’ external financing activities through bonds, notes, and other long-term borrowing arrangements. Depreciation, depletion, and amortization (DPACT) mea-

sures the total non-cash charges against earnings reflecting the allocation of tangible and intangible asset costs over their useful lives, as well as the depletion of natural resources. This variable captures the systematic expensing of capital investments and provides insight into the firm’s asset base and capital intensity. We construct the Depreciation Coverage Moderation signal by computing the year-over-year change in long-term debt issuance scaled by lagged depreciation expense: $(DLTIS(t) - DLTIS(t-1)) / DPACT(t-1)$. This ratio captures the acceleration or deceleration in debt financing relative to the firm’s existing depreciation base, providing a measure of how aggressively firms are changing their debt issuance patterns relative to their capital structure. The use of lagged depreciation as a scaling variable normalizes the change in debt issuance by the firm’s existing asset intensity, making the signal comparable across firms of different sizes and capital structures. We calculate this signal using fiscal year-end values and require non-missing values for both DLTIS and DPACT in consecutive years. To ensure meaningful ratios, we exclude observations where the lagged depreciation value is zero or negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCM signal. Panel A plots the time-series of the mean, median, and interquartile range for DCM. On average, the cross-sectional mean (median) DCM is -2.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCM data. The signal’s interquartile range spans -0.42 to 0.44. Panel B of Figure 1 plots the time-series of the coverage of the DCM signal for the CRSP universe. On average, the DCM signal is available for 5.82% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does DCM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCM portfolio and sells the low DCM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DCM strategy earns an average return of 0.21% per month with a t-statistic of 3.04. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.26% per month and have t-statistics exceeding 2.44 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.24 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 473 stocks and an average market capitalization of at least \$1,672 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-27bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.45. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the DCM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCM, as well as average returns and alphas for long/short trading DCM strategies within each size quintile. Panel

B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCM strategy achieves an average return of 32 bps/month with a t-statistic of 3.51. Among these large cap stocks, the alphas for the DCM strategy relative to the five most common factor models range from 26 to 35 bps/month with t-statistics between 2.83 and 3.92.

5 How does DCM perform relative to the zoo?

Figure 2 puts the performance of DCM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCM strategy falls in the distribution. The DCM strategy’s gross (net) Sharpe ratio of 0.42 (0.31) is greater than 83% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCM strategy (red line).² Ignoring trading costs, a \$1 invested in the DCM strategy would have yielded \$2.21 which ranks the DCM strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCM strategy would have yielded \$1.25 which ranks the DCM strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCM relative to those. Panel A shows that the DCM strategy gross alphas fall between the 51 and 62 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCM strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCM ranks between the 66 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCM with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCM or at least to weaken the power DCM has predicting the cross-section of returns. Figure 7 plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCM signal in these Fama-MacBeth regressions exceed 1.91, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DCM is 1.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCM strategy earns alphas that range from 14-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03,

which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCM trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.01.

7 Does DCM add relative to the whole zoo?

Finally, we can ask how much adding DCM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCM grows to \$1194.95.

8 Conclusion

This study provides compelling evidence that Depreciation Coverage Moderation (DCM) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCM possesses economically significant predictive

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCM is available.

power that persists even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short trading strategy based on DCM generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.42 and a net Sharpe ratio of 0.31 after accounting for transaction costs. The signal’s robustness is particularly noteworthy, as it delivers statistically significant abnormal returns of 17 basis points per month (t -statistic = 2.44) relative to the Fama-French five-factor model augmented with momentum. Even more impressively, DCM maintains its predictive power when confronted with a comprehensive set of controls, generating a monthly alpha of 14 basis points (t -statistic = 2.01) after controlling for six established factors plus six closely related strategies from the factor zoo, including net debt financing, inventory growth, and asset growth measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. DCM appears to capture a distinct dimension of firm characteristics related to capital allocation efficiency and financial health that is not fully subsumed by existing factors. The signal’s ability to generate positive risk-adjusted returns even after transaction costs suggests potential implementability in real-world portfolio construction, though practitioners should carefully consider market impact and capacity constraints.

Despite these promising results, several limitations warrant acknowledgment. First, while our analysis follows best practices for factor validation, the true out-of-sample performance of DCM remains to be observed as the signal becomes more widely known. Second, our study focuses primarily on U.S. equity markets, and the international robustness of DCM requires further investigation. Third, the economic mechanisms underlying the DCM anomaly deserve deeper theoretical and empirical exploration to better understand whether it represents a risk-based phenomenon or a behavioral mispricing.

Future research should pursue several promising avenues. Examining the performance of DCM across different market regimes and economic cycles would provide valuable insights into its stability and conditional effectiveness. Additionally, investigating the interaction between DCM and corporate events such as mergers, acquisitions, or financial distress could reveal important boundary conditions. Researchers should also explore whether the signal’s effectiveness varies across industries or firm characteristics such as size, age, or financial constraints. Finally, developing a more comprehensive theoretical framework linking depreciation coverage policies to expected returns would enhance our understanding of this anomaly and potentially uncover related predictive signals.

In conclusion, Depreciation Coverage Moderation emerges as a robust and economically meaningful predictor of stock returns that contributes incrementally to our understanding of the cross-section of expected returns. While further validation and theoretical development are warranted, this signal represents a valuable addition to the toolkit of quantitative investors and enriches our knowledge of the determinants of asset prices.

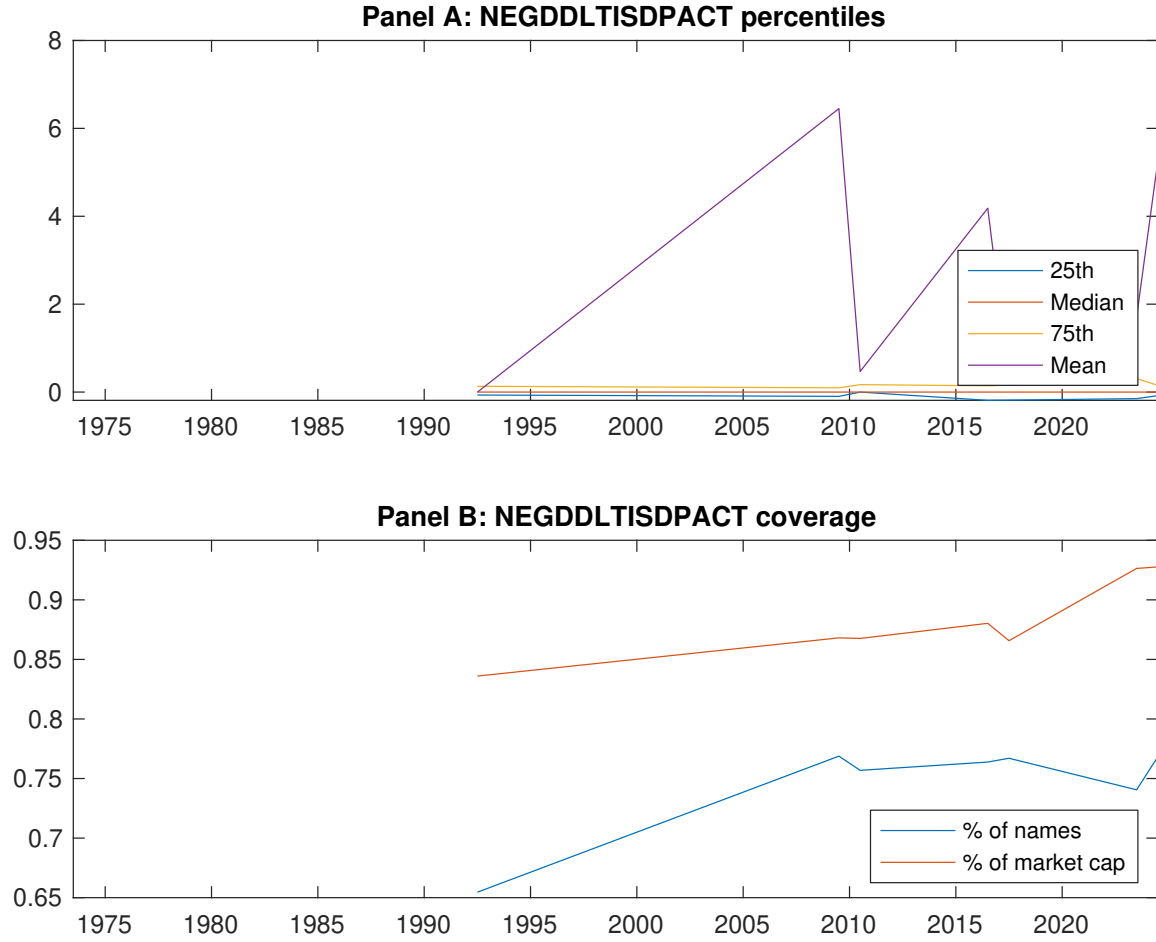


Figure 1: Times series of DCM percentiles and coverage. This figure plots descriptive statistics for DCM. Panel A shows cross-sectional percentiles of DCM over the sample. Panel B plots the monthly coverage of DCM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.66 [3.00]	0.63 [3.63]	0.65 [3.31]	0.69 [4.00]	0.87 [4.25]	0.21 [3.04]
α_{CAPM}	-0.09 [-1.70]	0.05 [0.92]	-0.01 [-0.16]	0.11 [2.17]	0.17 [3.22]	0.26 [3.83]
α_{FF3}	-0.08 [-1.49]	0.02 [0.48]	0.05 [0.96]	0.11 [2.18]	0.19 [3.59]	0.26 [3.82]
α_{FF4}	-0.06 [-1.08]	0.03 [0.55]	0.10 [1.72]	0.08 [1.53]	0.17 [3.24]	0.23 [3.26]
α_{FF5}	-0.05 [-0.99]	-0.06 [-1.21]	0.08 [1.47]	0.02 [0.40]	0.14 [2.65]	0.19 [2.75]
α_{FF6}	-0.04 [-0.75]	-0.05 [-1.02]	0.12 [2.04]	0.00 [0.06]	0.13 [2.48]	0.17 [2.44]
Panel B: Fama and French (2018) 6-factor model loadings for DCM-sorted portfolios						
β_{MKT}	1.11 [89.00]	0.95 [83.22]	0.97 [73.04]	0.94 [82.99]	1.05 [85.28]	-0.06 [-3.40]
β_{SMB}	0.09 [4.99]	-0.12 [-6.87]	-0.01 [-0.67]	-0.05 [-3.18]	0.15 [8.10]	0.06 [2.32]
β_{HML}	-0.01 [-0.24]	0.06 [2.67]	-0.17 [-6.55]	-0.07 [-3.31]	-0.12 [-5.05]	-0.11 [-3.63]
β_{RMW}	0.03 [1.34]	0.16 [7.13]	-0.03 [-1.06]	0.08 [3.66]	0.06 [2.59]	0.03 [0.94]
β_{CMA}	-0.14 [-3.94]	0.10 [3.04]	-0.05 [-1.36]	0.23 [6.96]	0.11 [2.96]	0.25 [5.24]
β_{UMD}	-0.02 [-1.61]	-0.01 [-1.24]	-0.05 [-3.90]	0.03 [2.31]	0.01 [1.01]	0.03 [1.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	473	989	520	645	
me (\$10 ⁶)	1700	2544	2113	2579	1672	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.04]	0.26 [3.83]	0.26 [3.82]	0.23 [3.26]	0.19 [2.75]	0.17 [2.44]
Quintile	NYSE	EW	0.17 [3.74]	0.19 [4.34]	0.18 [4.07]	0.18 [3.83]	0.17 [3.69]	0.17 [3.60]
Quintile	Name	VW	0.23 [3.19]	0.28 [3.82]	0.28 [3.80]	0.25 [3.34]	0.23 [3.10]	0.21 [2.85]
Quintile	Cap	VW	0.23 [3.54]	0.27 [4.32]	0.27 [4.24]	0.24 [3.64]	0.20 [3.08]	0.18 [2.75]
Decile	NYSE	VW	0.34 [3.65]	0.40 [4.30]	0.37 [4.04]	0.33 [3.51]	0.26 [2.78]	0.23 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.19]	0.20 [2.85]	0.20 [2.81]	0.17 [2.52]	0.13 [1.91]	0.12 [1.67]
Quintile	NYSE	EW	-0.09 [-1.45]					
Quintile	Name	VW	0.17 [2.35]	0.22 [2.99]	0.22 [2.95]	0.20 [2.72]	0.18 [2.39]	0.16 [2.20]
Quintile	Cap	VW	0.18 [2.72]	0.23 [3.57]	0.22 [3.47]	0.20 [3.16]	0.16 [2.50]	0.14 [2.27]
Decile	NYSE	VW	0.27 [2.87]	0.33 [3.50]	0.30 [3.25]	0.28 [2.99]	0.20 [2.20]	0.19 [2.02]

Table 3: Conditional sort on size and DCM

This table presents results for conditional double sorts on size and DCM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCM and short stocks with low DCM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCM Quintiles					DCM Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.64 [2.19]	0.93 [3.44]	0.95 [3.40]	1.06 [3.60]	0.74 [2.56]	0.10 [1.25]	0.13 [1.60]	0.11 [1.29]	0.08 [0.94]	0.11 [1.35]	0.09 [1.11]
	(2)	0.77 [2.79]	0.88 [3.57]	0.82 [3.26]	0.92 [3.76]	0.85 [3.30]	0.08 [1.04]	0.12 [1.55]	0.09 [1.16]	0.11 [1.43]	0.06 [0.77]	0.08 [1.03]
	(3)	0.79 [3.10]	0.82 [3.74]	0.80 [3.35]	0.84 [3.80]	0.93 [3.82]	0.14 [1.75]	0.17 [2.14]	0.17 [2.05]	0.15 [1.77]	0.17 [2.00]	0.15 [1.80]
	(4)	0.73 [3.15]	0.82 [3.96]	0.84 [3.77]	0.79 [3.82]	0.85 [3.86]	0.12 [1.64]	0.16 [2.11]	0.13 [1.76]	0.10 [1.32]	0.10 [1.30]	0.08 [1.01]
	(5)	0.55 [2.71]	0.55 [3.19]	0.61 [3.24]	0.60 [3.35]	0.87 [4.47]	0.32 [3.51]	0.35 [3.84]	0.35 [3.92]	0.30 [3.28]	0.30 [3.24]	0.26 [2.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCM Quintiles					DCM Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	366	368	368	368	363	33	30	29	29	32	
	(2)	101	102	101	102	101	55	56	55	55	55	
	(3)	72	73	72	73	72	100	99	96	98	99	
	(4)	61	62	62	62	61	218	226	218	223	215	
(5)	57	57	57	57	57	1451	1936	1684	2047	1467		

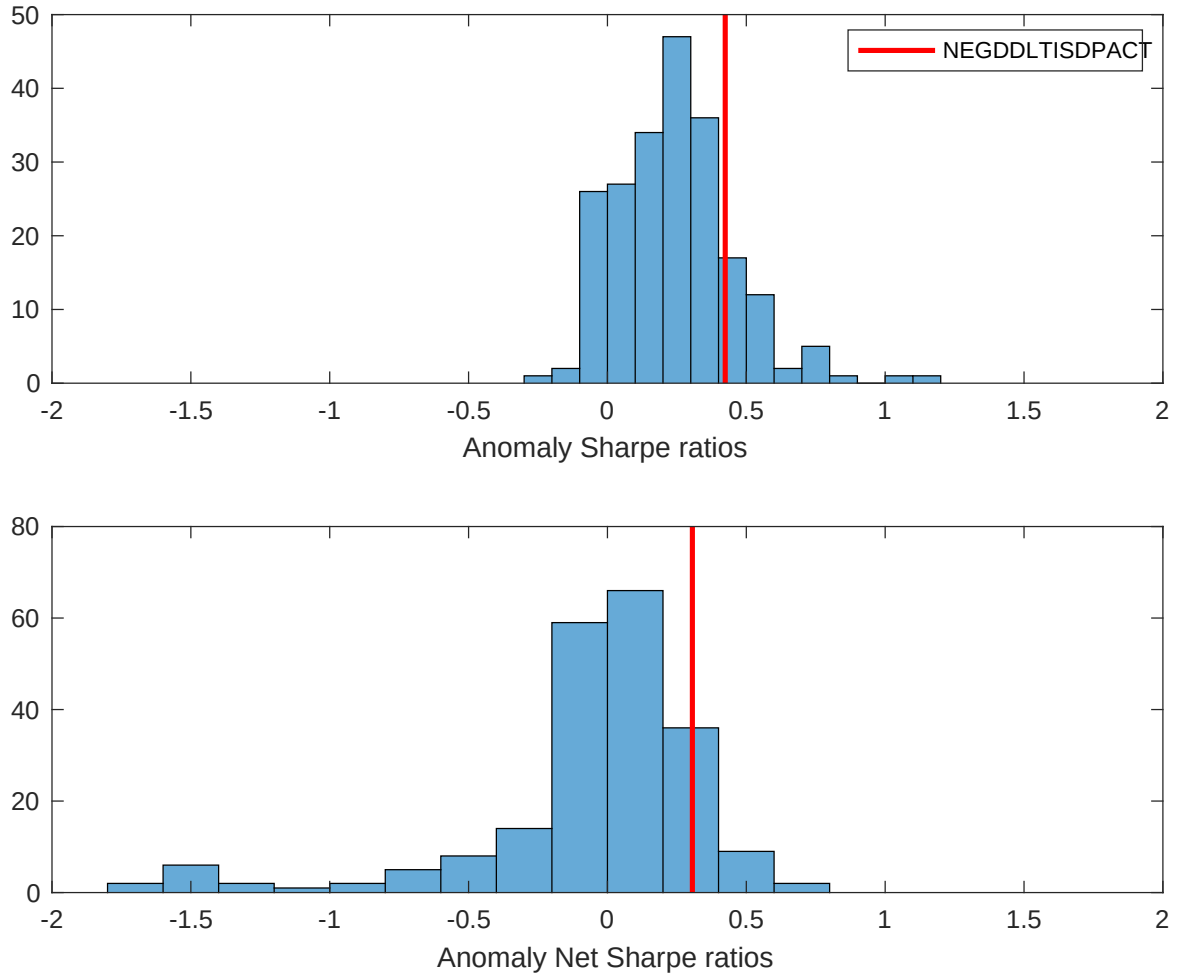


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

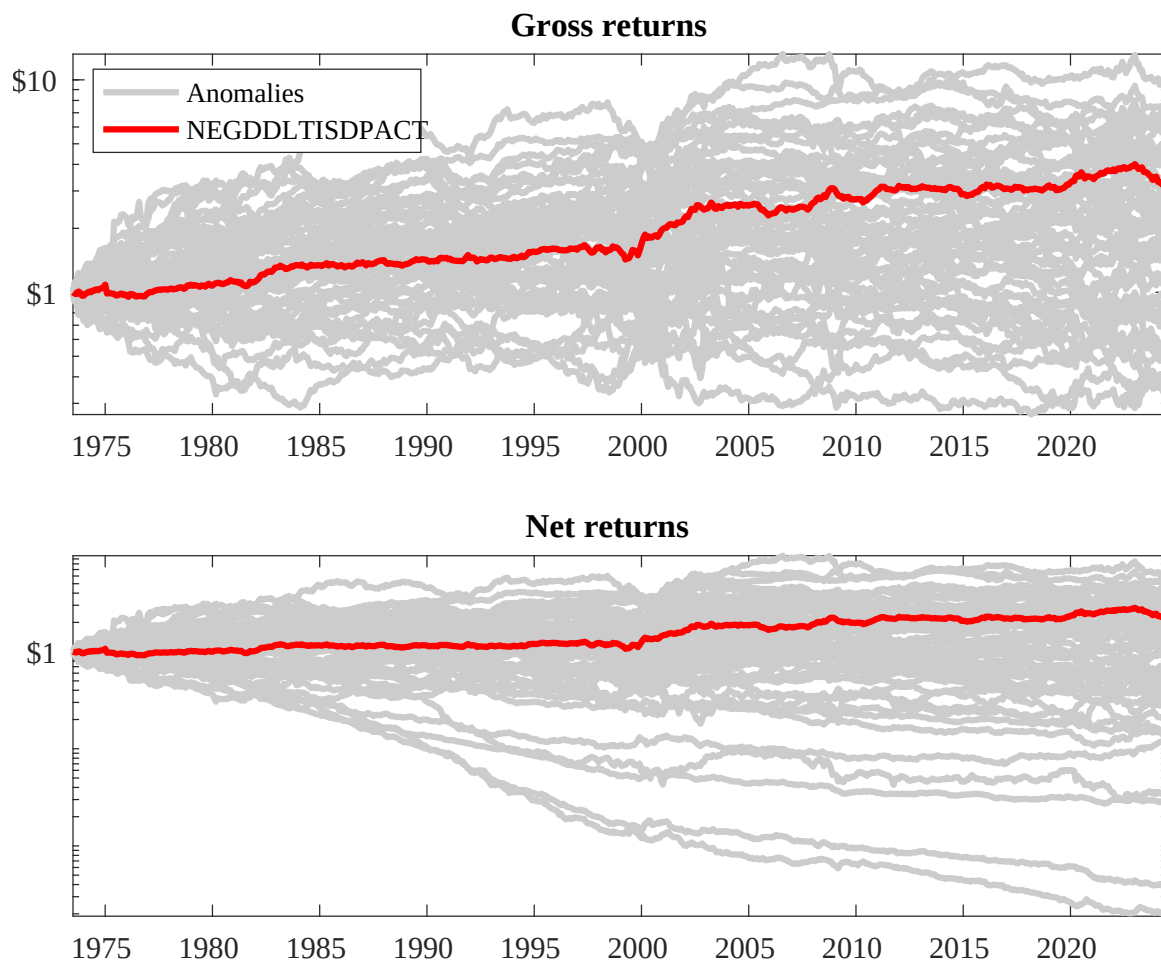
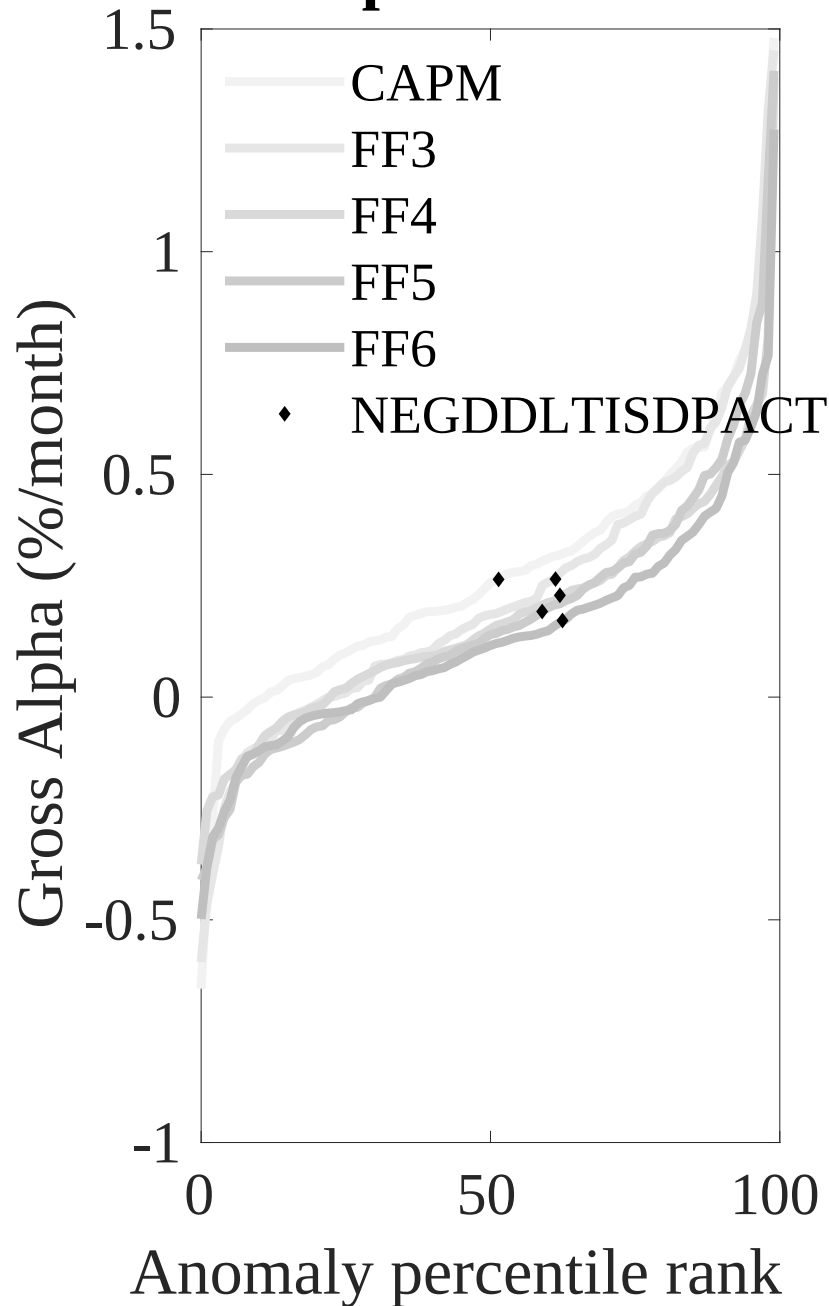


Figure 3: Dollar invested.

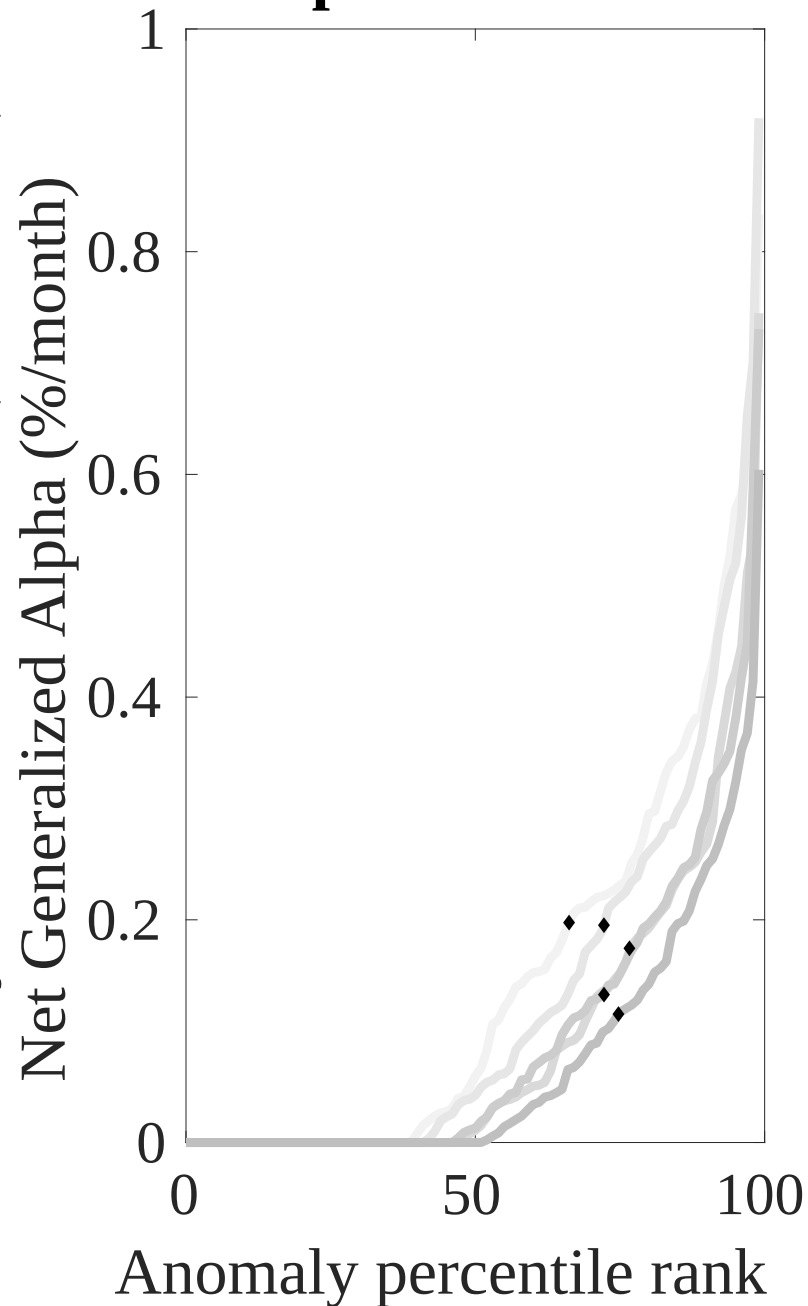
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



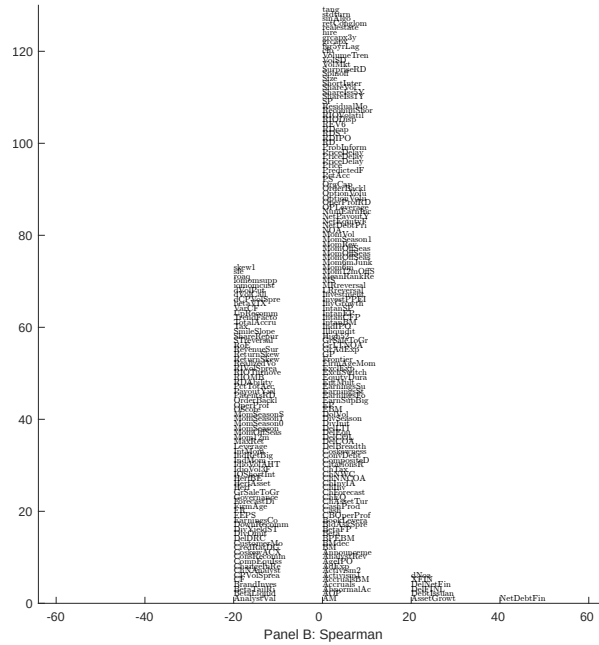
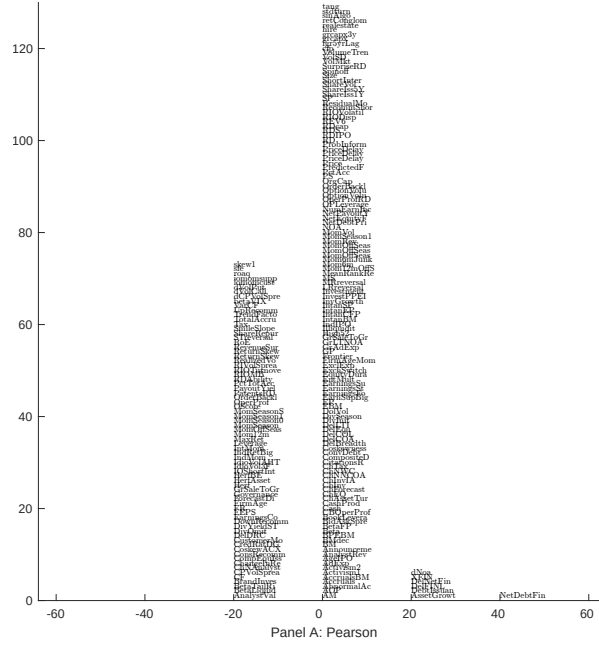


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DCM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

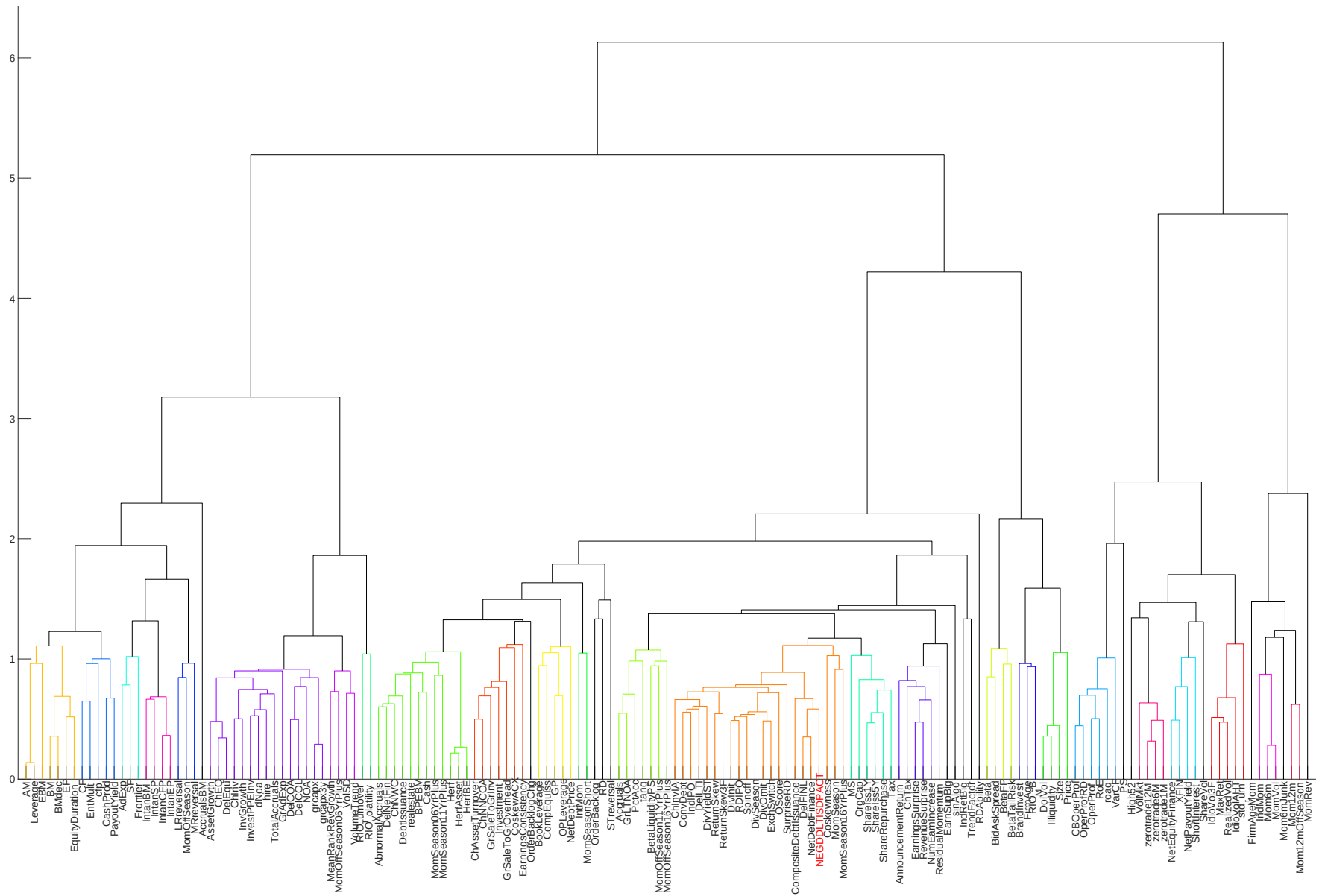


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

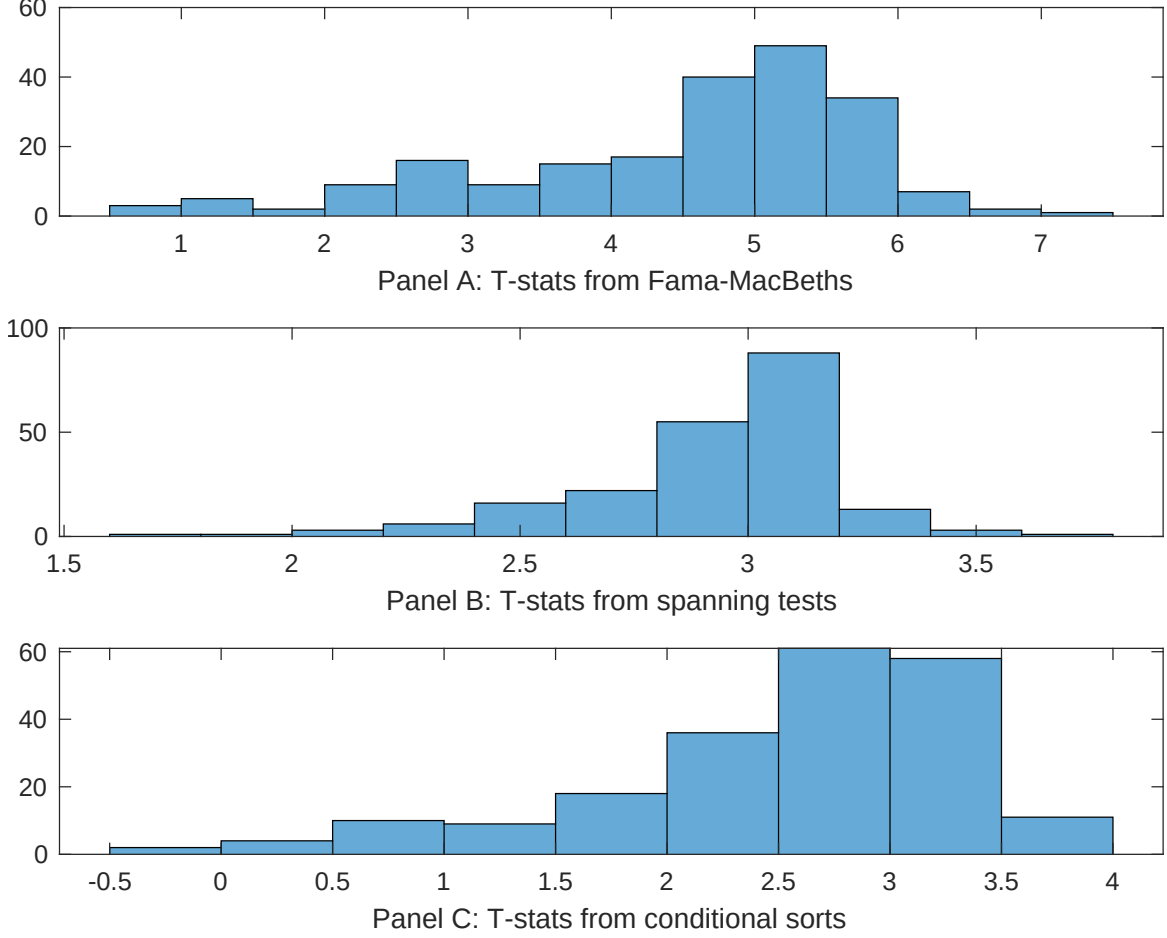


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.13 [5.38]	0.14 [5.72]	0.14 [5.30]	0.14 [5.79]	0.14 [5.76]	0.15 [5.74]
DCM	0.21 [2.65]	0.19 [2.78]	0.24 [3.49]	0.36 [4.28]	0.15 [1.91]	0.22 [3.01]	0.11 [1.18]
Anomaly 1	0.19 [8.72]						0.90 [1.32]
Anomaly 2		0.17 [9.20]					-0.80 [-1.66]
Anomaly 3			0.19 [6.35]				0.10 [1.80]
Anomaly 4				0.41 [6.95]			0.20 [0.34]
Anomaly 5					0.10 [8.29]		0.50 [2.33]
Anomaly 6						0.17 [7.58]	0.88 [3.14]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [2.09]	0.15 [2.10]	0.14 [2.03]	0.16 [2.32]	0.16 [2.26]	0.16 [2.27]	0.14 [2.01]
Anomaly 1	18.72 [4.99]						15.86 [3.08]
Anomaly 2		10.84 [2.75]					-0.78 [-0.15]
Anomaly 3			14.29 [4.13]				12.05 [3.19]
Anomaly 4				6.09 [2.25]			6.29 [2.22]
Anomaly 5					-0.72 [-0.18]		-6.87 [-1.64]
Anomaly 6						4.05 [1.29]	-1.73 [-0.52]
mkt	-5.17 [-3.28]	-4.96 [-3.10]	-3.24 [-1.96]	-5.32 [-3.32]	-5.22 [-3.24]	-5.27 [-3.28]	-3.62 [-2.18]
smb	4.42 [1.83]	4.96 [2.02]	10.66 [4.03]	6.74 [2.76]	6.13 [2.50]	6.09 [2.50]	9.80 [3.51]
hml	-10.19 [-3.35]	-10.22 [-3.30]	-9.36 [-3.03]	-11.13 [-3.60]	-10.88 [-3.50]	-11.37 [-3.64]	-8.73 [-2.82]
rmw	1.07 [0.34]	2.16 [0.68]	-5.41 [-1.45]	4.01 [1.27]	3.13 [0.99]	3.21 [1.02]	-4.89 [-1.29]
cma	20.29 [4.30]	21.51 [4.42]	15.84 [3.07]	19.73 [3.74]	25.99 [3.82]	21.92 [4.12]	17.56 [2.52]
umd	2.23 [1.41]	2.51 [1.54]	3.34 [2.12]	2.82 [1.74]	3.44 [2.15]	3.44 [2.15]	1.61 [0.99]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	11	12	10	10	10	15

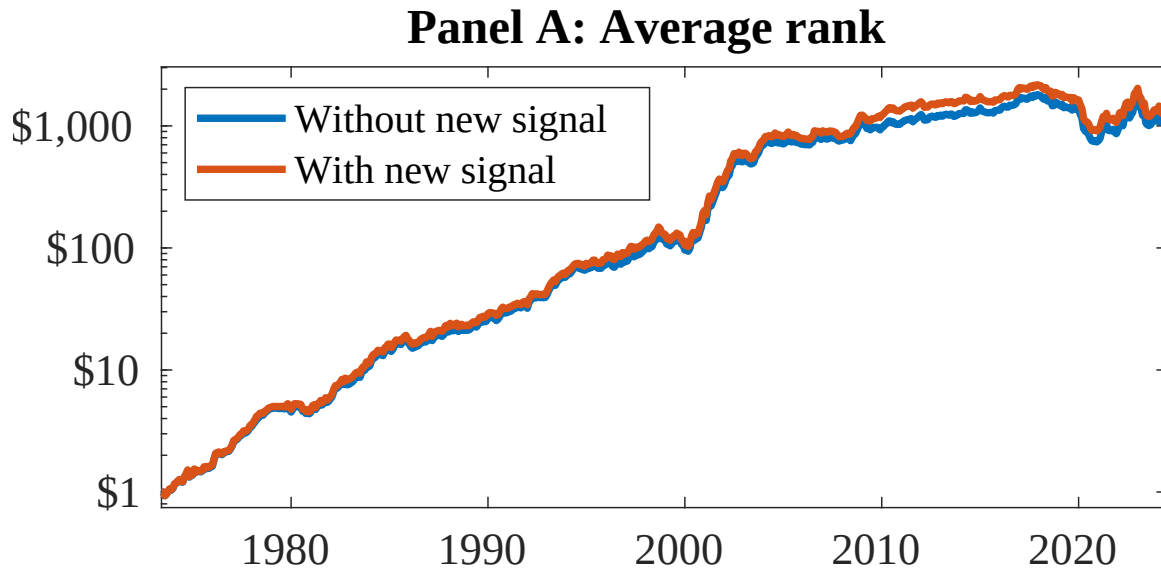


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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